

For the use of registered medical practitioner only

RANFUZOSIN

(Alfuzosin Hydrochloride Modified Release Tablets 10mg)

COMPOSITION

Each modified release tablet contains:

Alfuzosin Hydrochloride10 mg

Inactive ingredients:

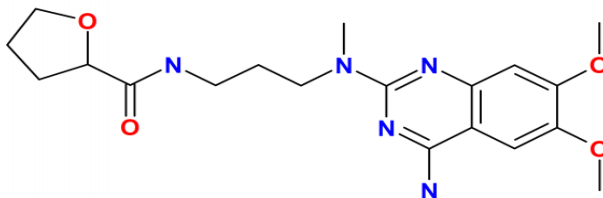
Lactose, Povidone, Aerosil, Purified Talc, Magnesium Stearate, Hypromellose, Hydroxy propyl cellulose

PRODUCT DESCRIPTION

White to off-white, uncoated, round, biconvex tablets debossed with 'RY10' on one side of the tablet.

DESCRIPTION

RANFUZOSIN is a modified release formulation that contains alfuzosin hydrochloride as active ingredient, which is a selective peripherally acting antagonist of post-synaptic α_1 -adrenoreceptors. Alfuzosin hydrochloride is chemically described as (R,S)-N-[3-[(4-amino-6,7-dimethoxy-2-quinazolinyl) methylamino] propyl] tetrahydro-2-furancarboxamide hydrochloride. The empirical formula of alfuzosin hydrochloride is $C_{19}H_{27}N_5O_4 \cdot HCl$. The molecular weight of alfuzosin hydrochloride is 425.9.



ALFUZOSIN

PHARMACODYNAMICS & PHARMACOKINETICS

Pharmacodynamics

Alfuzosin is an orally active quinazoline derivative. It is a selective peripherally acting antagonist of post-synaptic alpha1-adrenoreceptors.

Selectivity of alfuzosin for the alpha-1-adrenoreceptors located in the prostate, bladder base and prostatic urethra have reported *in vitro*.

Clinical manifestations of Benign Prostatic Hypertrophy (BPH) are associated with infra vesical obstruction which is triggered by both anatomical (static) and functional (dynamic) factors. The functional component of obstruction arises from the tension of prostatic smooth muscle which is mediated by alpha-adrenoceptors. Activation of alpha-1-adrenoceptors stimulates smooth muscle contraction, thereby increasing the tone of the prostate, prostatic capsule, prostatic urethra and bladder base, and, consequently, increasing the resistance to bladder outflow. This in turn leads to outflow obstruction and possible secondary bladder instability.

Alpha-blockade decreases infra vesical obstruction via a direct action on prostatic smooth muscle.

It has been reported that alfuzosin decreases urethral pressure and therefore, resistance to urine flow during micturition in animal models. Moreover, alfuzosin inhibits the hypertonic response of the urethra more readily than that of vascular muscle and shows functional uroselectivity in conscious normotensive rats by decreasing urethral pressure at doses that do not affect blood pressure.

Alfuzosin improves voiding parameters in man by reducing urethral tone and bladder outlet resistance, and facilitates bladder emptying.

Significant increase in urine flow, significant decrease in detrusor pressure, significant increase the volume producing the sensation of the need to urinate and significant reduced residual urine volume have been reported in BPH patients receiving alfuzosin. These favourable urodynamic effects result in an improvement in the irritant and obstructive urinary symptoms.

Alfuzosin may cause moderate antihypertensive effects.

A lower frequency of acute urinary retention has been reported in patients receiving alfuzosin than untreated patients.

Pharmacokinetics

Alfuzosin HCl is approximately 90% bound to plasma proteins. Alfuzosin is extensively metabolized by the liver with excretion in the urine of only 11% of the unchanged compound. The majority of metabolites (which are inactive) are excreted in the faeces (75-90%). The pharmacokinetic profile of alfuzosin is not modified in the case of chronic cardiac insufficiency.

Modified release formulation: The mean value of relative bioavailability is 104.4% after administration of a dose of 10 mg, compared with that of the immediate-release formulation at a dosage of 7.5 mg (25 mg 3 times daily) in healthy volunteers of average age. The peak plasma concentration is reached 9 hrs after administration as against 1 hr for the immediate formulation.

The apparent elimination half-life is 9.1 hrs. Bioavailability is increased when the product is administered after a meal (see DOSE/USE INSTRUCTION).

The mean values of $C_{\max}$ and AUC are moderately increased in patients with moderate impairment of renal function (creatinine clearance >30 mL/min) without effect on the elimination half-life compared with patients with normal renal function. Dosage adjustment is not necessary in patients with impaired renal function with creatinine clearance >30 mL/min.

Special Populations

Geriatric: The pharmacokinetic properties ($C_{\max}$ and AUC) are not increased in elderly patients, compared with healthy volunteers of average age.

Renal Insufficiency

Relative to subjects with normal renal function (CL_{CR} >80 mL/min), the mean $C_{\max}$ and AUC values were increased by approximately 50% in patients with mild (CL_{CR} 60 to 80 mL/min), moderate (CL_{CR} 30 to 59 mL/min), or severe (CL_{CR} 30 mL/min), renal impairment. (See: WARNINGS AND PRECAUTIONS).

Hepatic Insufficiency

In clinical studies conducted by the innovator, in patients with moderate or severe hepatic insufficiency, plasma apparent clearance of alfuzosin hydrochloride was found to be reduced to about 1/3rd to 1/4th of that observed in healthy subjects. Therefore, alfuzosin is contraindicated in patients with hepatic impairment. In mild hepatic insufficiency, the pharmacokinetics of alfuzosin hydrochloride has not been studied.

INDICATIONS / USAGE

Treatment of functional symptoms of benign prostatic hypertrophy. Adjuvant treatment to a catheter in acute urinary retention related to benign prostatic hypertrophy.

DOSE/USE INSTRUCTION

The tablet must be swallowed whole with a glass of water.

The recommended dose is one 10mg tab daily to be taken immediately after the evening meal.

Adjuvant treatment to a catheter in acute urinary retention related to benign prostatic hypertrophy: The recommended dosage is one 10 mg tab/day, to be taken after a meal, from the 1st day of catheterization onwards. The treatment is administered for 3-4 days, i.e. 2-3 days while the catheter is in place and 1 day after it is removed.

CONTRAINDICATIONS

Hypersensitivity to alfuzosin hydrochloride or any of excipients

History of orthostatic hypotension

Combination with other alpha-blockers

Hepatic insufficiency

Severe renal insufficiency (creatinine clearance <30 mL/min)

Use with potent CYP3A4 inhibitors such as ketoconazole, itraconazole and ritonavir, since it increases alfuzosin levels (see DRUG INTERACTIONS).

WARNINGS AND PRECAUTIONS

- **General**

Prostatic Carcinoma: Carcinoma of the prostate and BPH cause many of the same symptoms. These two diseases frequently coexist. Therefore, patients thought to have BPH should be examined prior to starting therapy with alfuzosin to rule out the presence of carcinoma of the prostate.

Intraoperative Floppy Iris Syndrome (IFIS) has been observed in some patients currently or previously taking alpha-1 blockers. It is characterized by the combination of a flaccid iris that billows in response to intraoperative irrigation currents, progressive intraoperative miosis despite preoperative dilation with standard mydriatic drugs, and potential prolapse of the iris towards the phacoemulsification incisions. As IFIS may lead to procedural complications during cataract complications, the operating ophthalmologist should be informed of the current or past use of alpha-1 blockers including alfuzosin. The ophthalmologist may have to modify the surgical technique, such as using iris hooks, iris dilator rings or viscoelastic substances. Stopping alpha-1 blocker therapy prior to cataract surgery does not appear to be of benefit.

Drug-Drug Interactions: The pharmacokinetic and pharmacodynamic interactions between alfuzosin and other alpha-blockers have not been determined. However, interactions may be expected, and alfuzosin should not be used in combination with other alpha-blockers.

Coronary Insufficiency: In patients with coronary insufficiency, anti-anginal therapy should be continued. However, if symptoms of angina pectoris newly develop, reappear or worsen, Ranfuzosin should be discontinued.

Hepatic Insufficiency: Alfuzosin should not be given to patients with moderate or severe hepatic insufficiency. (See: Contraindications). The pharmacokinetics of alfuzosin have not been studied in patients with mild hepatic insufficiency

In patients with severe renal impairment (creatinine clearance below 30 mL/min), the experience with the use of alfuzosin is limited, therefore caution, with the use of Ranfuzosin is advised in these patients.

In patients with hypersensitivity to alpha-1 blockers, treatment with Ranfuzosin should be started gradually. Ranfuzosin should be administered carefully to patients being treated with antihypertensives. Blood pressure should be monitored regularly, especially at the beginning of treatment.

Patients with Congenital or Acquired QT Prolongation: In a study conducted by the innovator, the QT effect appeared less with alfuzosin 10 mg than with 40 mg, and the effect of alfuzosin 40 mg did not appear as large as that of the active control moxifloxacin at its therapeutic dose. A post-marketing study conducted by the innovator evaluating the effect of combining alfuzosin with another drug of comparable QT effect showed an increased effect when compared to either drug alone. The QT increase with both drugs was no more than additive and was lower than that of the active control moxifloxacin. These observations should be considered in clinical decisions when prescribing alfuzosin for patients with a known history of QT prolongation or patients who are taking medications which prolong the QT interval. There has been no signal of Torsades de Pointe in the extensive post-marketing experience with alfuzosin. There are no known PK/PD studies of the effects of other alpha-blockers on cardiac repolarization.

Ranfuzosin contains 77 mg lactose per tablet. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

- **Information for Patients**

Patients should be told about the possible occurrence of symptoms related to postural hypotension, such as dizziness, when beginning alfuzosin, and they should be cautioned about driving, operating machinery, or performing hazardous tasks during this period.

Alfuzosin should be taken with food and with the same meal each day. Patients should be warned that Ranfuzosin tablet should be swallowed whole. Any other mode of intake such as crushing, crunching, chewing, grinding or pounding to powder should be prohibited. These actions may lead to inappropriate release and absorption of the drug and therefore possible early adverse reactions.

- **Laboratory Tests**

No laboratory test interactions with alfuzosin tablets are known.

- **Warnings**

Postural hypotension with or without symptoms (e.g., dizziness) may develop within a few hours following administration of alfuzosin. As with other alpha-blockers, there is a potential for syncope. Patients should be warned of the possible occurrence of such events and should avoid situations where injury could result should syncope occur. There may be an increased risk of hypotension/postural hypotension and syncope when taking alfuzosin concomitantly with anti-hypertensive medication or nitrates. Care should be taken when alfuzosin is administered to patients with symptomatic hypotension or patients who have had a hypotensive response to other medications. These effects do not usually require discontinuation of treatment and are transient in nature. Dose adjustment may, however, be required. Blood pressure should be monitored regularly especially at the beginning of the treatment with Ranfuzosin.

The patient experiencing postural hypotension should be advised to lie down until improvement in symptoms takes place. He should be suitably informed and instructed to avoid situations in which he might get injured should syncope occur.

DRUG INTERACTIONS

Inadvisable Combinations

- Alpha-1- receptor blockers (prazosin, urapidil, minoxidil): Enhanced hypotensive effect. Risk of severe orthostatic hypotension (See CONTRAINDICATIONS).
- Ketoconazole, itraconazole, ritonavir, clarithromycin, erythromycin: Risk of increased plasma alfuzosin concentrations and increased undesirable effects (See CONTRAINDICATIONS).

Combination to be taken into account:

- Antihypertensive agents: Increased antihypertensive effect and risk of orthostatic hypotension (additive effect).
- Nitrates (see WARNINGS AND PRECAUTIONS).
- *PDE5 Inhibitors*: Caution is advised when alpha adrenergic blocking agents, including alfuzosin, are co-administered with PDE5 inhibitors. Alpha-adrenergic blockers and PDE5 inhibitors are both vasodilators that can lower

blood pressure. Concomitant use of these two drug classes can potentially cause symptomatic hypotension.

The administration of general anaesthetics to patients on alfuzosin can cause profound hypotension. It is recommended that the tablet should be withdrawn 24 hours before surgery.

Other interactions

No pharmacodynamic or pharmacokinetic interaction has been reported in healthy volunteers between alfuzosin and the following drugs: warfarin, digoxin, hydrochlorothiazide and atenolol.

ADVERSE REACTIONS / SIDE EFFECTS

Expected frequencies are classified as under:

Very Common ($>1/10$), common ($>1/100$ to $<1/10$), uncommon ($>1/1,000$ to $<1/100$), rare ($>1/10,000$ to $<1/1000$), and very rare ($<1/10,000$) including isolated reports.

Nervous System:

Common: faintness/dizziness, headache

Uncommon: vertigo, malaise, drowsiness

Cardiovascular System:

Uncommon: tachycardia, palpitations, hypotension (postural), syncope

Very rare: New Onset, aggravation or recurrence of angina pectoris in patients with pre-existing coronary artery disease

Gastro-intestinal System:

Common: nausea, abdominal pain,

Uncommon: diarrhoea, dry mouth

Skin and Subcutaneous Tissue Disorders:

Uncommon: rash, pruritus

Very rare: urticaria, angioedema

General Disorders:

Common: asthenia,

Uncommon: flushes, edema, chest pain,

Respiratory System:

Common: bronchitis, sinusitis, pharyngitis

In addition to the above following adverse events have also been reported: flushing, dyspepsia, constipation, pain, impotence, priapism, and rhinitis.

Post-Marketing Adverse Event Reports:

The following adverse reactions have been identified during post approval use of alfuzosin. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency.

General disorders: edema

Cardiac disorders: tachycardia, chest pain, angina pectoris in patients with pre-existing coronary artery disease

Gastrointestinal disorders: diarrhea Hepatobiliary disorders: hepatocellular and cholestatic liver injury (including cases with jaundice leading to drug discontinuation)

Respiratory system disorders: rhinitis

Reproductive system disorders: priapism

Skin and subcutaneous tissue disorders: rash, pruritis, urticaria, angioedema

Vascular disorders: flushing

During cataract surgery, a variant of small pupil syndrome known as Intraoperative Floppy Iris Syndrome (IFIS) has been reported in some patients on or previously treated with alpha-1 blockers (see **PRECAUTIONS**).

PREGNANCY AND LACTATION

Alfuzosin is not indicated for use in women. The safety of alfuzosin during the passage in the breast milk is not known.

OVERDOSE AND TREATMENT

In case of overdose of alfuzosin, maintenance of blood pressure is of prime importance, as it may present as hypotension. Keeping the patient in supine position can help restore blood pressure and normalize heart rate. Administration of intravenous fluids should be considered. Vasopressors may be required. Monitor renal function and give support as needed. Dialysis may not benefit as alfuzosin is 82-90% protein bound.

STORAGE

Store below 30°C, Protected from light and moisture

KEEP ALL MEDICINES OUT OF REACH OF CHILDREN

PACKING

Blister pack of 10's. Pack of 3 x 10's

Date of revision: April 2017

Product Registration Holder / Manufactured by:

RANBAXY (MALAYSIA) SDN. BHD.

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